

Miqdad Hayati

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EDUCATION

- **Habib University** Karachi, Pakistan
Bachelor of Science - Computer Science; GPA: 3.50/4.00 *Aug 2024 - Jun 2028*
- **A-Levels:** 1 A, 3 B
- **O-Levels:** 3 A*, 4 A, 2 B

SKILLS SUMMARY

- **Programming Languages:** Python, C++, Verilog HDL, JavaScript, SQL, HTML, CSS
- **Hardware & Embedded Systems:** FPGA Development (Basys-3), VGA Signal Processing, Sensor Interfacing (LDR / Optical), Hardware-Software Co-Design
- **Frameworks & Libraries:** SFML (Simple and Fast Multimedia Library), NumPy, Matplotlib, Pandas, Playwright, BeautifulSoup4
- **Databases:** MySQL, NoSQL, Postgres
- **Concepts & Methodologies:** Object-Oriented Programming (OOP), Design Patterns (State, Strategy, Inheritance, Polymorphism), Data Structures & Algorithms, Metric Trees / VP-Trees, Computational Modeling, Vector Calculus
- **Developer Tools:** Vivado, Git / GitHub, VS Code, LaTeX

EXPERIENCE

- **IBEX Pakistan** Onsite
Customer Service Specialist *May 2025 - Jul 2025*
 - Handled high-volume inbound customer interactions for a Tier-1 BPO client, resolving service inquiries while consistently meeting quality-assurance and average-handle-time targets.
 - Diagnosed and de-escalated complex customer issues across billing, technical, and account-related domains, coordinating cross-functionally with internal teams to drive first-contact resolution.
 - Maintained accurate case documentation in CRM tooling and contributed to knowledge-base entries that improved resolution efficiency for recurring inquiry patterns.

RESEARCH PUBLICATIONS

- **Electromagnetic Vector Fields: Mathematical Foundations, Analytical Modeling, and Computational Optimization of Uniform Magnetic Field Generators** - *Co-Author, Dec 2025*
 - Authored a rigorous technical report applying vector calculus and Maxwell's equations to derive the analytical conditions for magnetic-field uniformity in Helmholtz coil systems, formally proving the optimal coil-separation criterion ($d = R$) via Taylor-series expansion and elimination of even-order derivatives.
 - Engineered a Python simulation of the Biot-Savart law using filamentary current discretization to compute the 3D magnetic flux density $\mathbf{B}$ across a meshgrid, validating the analytical model with a Matplotlib streamplot visualization that confirmed the predicted central "volume of uniformity."

PROJECTS

- **Hardware-Based Duck Hunt Engine & Optical Targeting System:** An end-to-end NES-style light-gun game implemented in Verilog HDL on the Basys-3 FPGA with native VGA output and a custom-built optical aiming peripheral.
 - **Architected** the complete game engine in Verilog, including finite-state machines for sprite rendering, target spawning, score tracking, and round progression, all driven through a 640×480 VGA pipeline on the Basys-3.
 - **Innovated** an aim-tracking system that replaced the high-speed photodiode of the original NES Zapper with a low-cost LDR (Light-Dependent Resistor), overcoming a critical hardware constraint and material-cost barrier.

- **Engineered** a physical pinhole light-focusing apparatus from a drinking straw to collimate incoming light onto the LDR, enabling reliable target detection on modern LCD/LED panels and eliminating the legacy CRT-monitor dependency.
 - **Result:** Delivered a fully playable retro-gaming peripheral that demonstrated hardware-software co-design and creative problem-solving under tight FPGA resource constraints.
- **Plagiarism Detection System via VP-Metric Trees:** A high-performance document-similarity engine for accelerated near-duplicate text retrieval. (*Python, Information Retrieval, Data Structures*)
 - **Engineered** a sentence-level vectorization pipeline that segmented documents and converted them into word-frequency feature vectors for fine-grained similarity analysis.
 - **Implemented** Vantage-Point (VP) Trees with cosine and Euclidean distance metrics, enabling sub-linear nearest-neighbor queries that significantly outperformed brute-force $O(n)$ similarity comparisons on large corpora.
 - **Designed** a sentence-level traceback mechanism that pinpoints the exact overlapping passages between source and query documents, moving beyond aggregate similarity scores toward actionable evidence highlighting.
 - **Developing** a visualization UI to display matched passages side-by-side with color-coded highlighting for academic review use cases.
- **“Defend the Keep” – 2D Wave-Based Tower Defense Game:** A real-time strategy game inspired by *Plants vs. Zombies*. (*C++, SFML, OOP Design Patterns*)
 - **Developed** the full game loop, rendering, and input pipeline in C++ using the SFML multimedia library, supporting smooth 60 FPS gameplay with multiple concurrent entity types.
 - **Applied** rigorous object-oriented design — inheritance, polymorphism, and the **State design pattern** — to model tower archetypes, enemy classes, and clean transitions between menu, gameplay, pause, and game-over states.
 - **Designed** the in-game economy and difficulty curve, including gold-based resource management, tiered tower abilities with distinct cooldowns and damage profiles, escalating wave intensity, and a final boss encounter as the climactic challenge.

HONORS AND AWARDS

- **Pre-University Performance** - *O-Levels: 3 A*, 4 A, 2 B; A-Levels: 1 A, 3 B*
- **Co-Author, Technical Research Report** *Bridging electromagnetism, vector calculus, and computational simulation* - Dec 2025
- **GPA** - 3.50 / 4.00 across CS coursework at Habib University